



Unit III: Motivation and Leadership (6 Hrs.)

- 3.1 Motivation
 - 3.1.1 Theories of motivation: Maslow's hierarchy, Herzberg's two factor, Expectancy, Equity
 - 3.1.2 Techniques for motivation
- 3.2 Leadership
 - 3.2.1 Leadership styles: autocratic, democratic, servant and transformational
 - 3.2.2 Characteristics of learning organization in the ICT industry
- 3.3 Challenges and strategies for motivating and leading technical workforce

Concept of Motivation

- **Motivation** is the psychological process that drives individuals to take action towards achieving specific goals.
- It encompasses the internal and external factors that stimulate desire and energy in people to be continually interested and committed to a task.
- For computer engineering students, motivation is critical in tackling complex projects like software development, system design, or debugging.
- It can stem from **intrinsic factors** (e.g., passion for coding, solving problems) or **extrinsic factors** (e.g., grades, job prospects).

Concept of motivation

- *The term motivation has been derived from the word **motive**, which means **urge** to do or not to do something.*
- *It is an act of inspiring or encouraging people working in organization for devoting their optimum effort.*

Motivation thus ...

- *Psychological process*
- *Complex and unpredictable*
- *Goal directed behaviour*
- *System orientation*
- *Perceived relationship between reward and effort*
- *Concerned with whole individual*

- **1. Based on Origin**

- a. Intrinsic Motivation

- *Comes from within the person.*
- *Driven by interest, enjoyment, personal satisfaction.*
- **Example:** *Studying because you love learning.*

- b. Extrinsic Motivation

- *Comes from outside forces.*
- *Driven by rewards or avoidance of punishment.*
- **Example:** *Studying to get good grades or avoid scolding.*

2. Based on Purpose

a. Positive Motivation

- *Encourages behavior through rewards or benefits.*
- **Example:** *Bonus for meeting sales targets.*

b. Negative Motivation

- *Encourages behavior by avoiding undesirable consequences.*
- **Example:** *Doing a task to avoid penalty.*

Need for Motivation

- **Improved Performance:**
 - Motivated individuals work harder and produce better results.
- **Goal Achievement:**
 - It aligns efforts toward achieving personal or organizational goals.
- **Employee Engagement:**
 - Motivation increases commitment, reducing turnover and absenteeism.
- **Innovation and Creativity:**
 - Motivated individuals are more likely to think creatively and take initiative.
- **Satisfaction and Morale:**
 - It enhances job satisfaction, leading to a positive work environment.
- **Adaptability:**
 - Motivated employees are more willing to embrace change and challenges.

3.2. Incentives

- Incentives are rewards or benefits designed to encourage specific behaviors or performance.
- incentives can be **monetary** (e.g., bonuses for meeting project deadlines, scholarships for academic excellence) or **non-monetary** (e.g., recognition in hackathons, open-source contributions boosting your GitHub profile).
- **For example**, tech companies often offer stock options, flexible work hours, or access to cutting-edge tools as incentives. As future engineers, you might encounter incentives like internship opportunities or certifications (e.g., Google Cloud) that motivate you to upskill. Understanding how incentives work can help you prioritize tasks and negotiate better career opportunities.

Financial and Non-Financial Motivation

Financial Motivation

- **Salary Increases:** Direct financial compensation for performance.
- **Bonuses:** Additional payments based on performance metrics.
- **Commission:** Earnings based on sales or performance levels.
- **Benefits:** Health insurance, retirement plans, and other monetary perks.



Non-Financial Motivation

- **Recognition:** Acknowledgment of achievements and contributions.
- **Career Development:** Opportunities for training, promotions, and skill enhancement.
- **Work Environment:** Creating a positive and supportive workplace culture.
- **Work-Life Balance:** Policies that support flexible working hours and personal time.



Theories of motivation:

- Maslow's hierarchy,
- Herzberg's two factor,
- Expectancy,
- Equity

1. Maslow's Hierarchy of Needs

- Abraham Maslow in 1943 AD
- Maslow proposed a five-tier model of human needs, often depicted as a pyramid:



- **Physiological Needs:** Basic survival needs (food, water, shelter).
- **Safety Needs:** Security, stability, and protection.
- **Social Needs:** Belongingness, love, and relationships.
- **Esteem Needs:** Self-respect, recognition, and status.
- **Self-Actualization:** Achieving one's full potential. Individuals must satisfy lower-level needs before moving to higher ones.

Strength

- Provides a clear, intuitive framework for understanding motivation.
- Broad applicability in contexts like workplace management (e.g., ensuring basic needs like salary before focusing on growth opportunities).
- Emphasizes the progression of needs, useful for long-term goal setting.

Weakness

- Assumes a strict hierarchy, but people may prioritize needs differently (e.g., pursuing esteem over safety in risky jobs).
- Lacks empirical evidence to support universal applicability across cultures.
- Ignores simultaneous pursuit of multiple needs (e.g., seeking social connection while still addressing physiological needs).

2. Dual Factor Theory (Herzberg's Two-Factor Theory)

- Proposed by **Frederick Herzberg 1950AD**
- this theory differentiates between factors that cause job satisfaction and those that cause dissatisfaction.
- This theory suggests that job satisfaction and dissatisfaction are influenced by two independent sets of factors
- **Factors:**
 - **Hygiene Factors:** (e.g., salary, work conditions) do not motivate but can cause dissatisfaction if inadequate.
 - **Motivators:** (e.g., achievement, recognition) lead to job satisfaction and motivate employees to perform better.
- **Implication:** Improving hygiene factors can prevent dissatisfaction, but true motivation comes from enhancing motivators.

1. Hygiene Factors:

- These are extrinsic factors related to the job context or environment.
- They do not necessarily motivate employees but can lead to dissatisfaction if they are absent or inadequate.
- Think of them as the basic conditions necessary to prevent unhappiness at work.
- **Examples include:**
 - *Salary and benefits*
 - *Company policies and administration*
 - *Supervision*
 - *Working conditions*
 - *Interpersonal relationships*
 - *Job security*
 - *Status*

2. Motivator Factors:

- These are intrinsic factors related to the job content itself.
- They can lead to job satisfaction and motivation when present.
- These factors fulfill an individual's need for growth, recognition, and a sense of achievement.
- **Examples include:**
 - *Achievement*
 - *Recognition*
 - *The work itself*
 - *Responsibility*
 - *Advancement*
 - *Growth*

Strength

- Separates motivators (e.g., recognition for coding) from hygiene factors (e.g., good tools) for practical motivation strategies
- Clear differentiation between satisfaction and dissatisfaction factors.
- Focus on job enrichment.
- Practical application in organizational settings.

• Weakness

- Oversimplifies by not accounting for overlapping effects, like a salary motivating beyond just preventing dissatisfaction
- Oversimplification of human motivation.
- Cultural differences in motivators.
- Lack of empirical support.

Expectancy Theory

- **Victor Vroom** introduced Expectancy Theory in 1964 in his book *Work and Motivation*.
- Vroom was a Canadian psychologist and professor at Yale School of Management, focusing on workplace motivation and decision-making.
- motivation depends on three interconnected beliefs:
- **Expectancy (Effort → Performance):**
 - The belief that effort will lead to successful performance. For computer engineering students, this might mean believing that dedicating hours to studying data structures will result in writing efficient algorithms for a project. If you doubt your ability (e.g., due to lack of resources or prior failures), your motivation decreases.
- **Instrumentality (Performance → Reward):**
 - The belief that performance will lead to a desired reward. For instance, if you successfully complete a machine learning project (performance), you might expect a high grade, a professor's recommendation, or a portfolio piece to impress recruiters (reward). If the reward isn't guaranteed (e.g., the professor doesn't value the project), motivation wanes.
- **Valence (Value of Reward):**
 - The personal value you place on the reward. If you highly value the reward—like landing a job at a top tech firm (e.g., Google) through a strong project—your motivation increases. But if the reward isn't appealing (e.g., you don't care about grades), you'll be less motivated.

Motivation = Expectancy × Instrumentality × Valence.

Strengths and Limitations:

Strength:

- It emphasizes individual perceptions, making it practical for personalized motivation strategies in tech teams.
- Individual focus on perceptions and expectations.
- Clear motivational framework.
- Useful for goal setting.

Limitation:

- It assumes rational decision-making, but emotions or external factors (e.g., team conflicts, tight deadlines) can influence motivation in real-world engineering projects.
- Complexity with multiple variables.
- Assumes rational decision-making.
- Limited scope regarding intrinsic motivations.

Adams' Equity Theory of Motivation (1963–1965)

- Propounded by **J. Stacy Adams**
- **Core Concept**
- Employees remain motivated only when they perceive **fairness** in the ratio of their inputs and outputs compared to others.
- Employees compare their input-output ratio with others
- Inputs: effort, skill, overtime, stress
- Outputs: salary, recognition, growth, perks
- If ratio feels unequal → tension & demotivation
- Person can reduce effort, demand more pay, or leave

Strength:

- Explains real-world attrition in Indian IT companies very accurately
- Justifies the need for **transparent salary bands** (like Google, Microsoft)
- Helps HR design fair appraisal & reward systems
- Explains office politics and “favouritism” complaints

Limitation:

- Perception of fairness is highly subjective (one person’s “hard work” is another’s “normal”)
- People tend to overestimate own inputs and underestimate others’ inputs
- Choosing referent other is personal – difficult for management to control
- Over-rewarded people rarely feel guilty for long
- Does not explain why some employees stay even when treated unfairly (family pressure, market condition)

Contemporary Issues of Motivation in Nepalese Organizations

- *Economic Constraints*
- *Cultural Influences*
- *Workforce Diversity*
- *Political Instability*
- *Technological Gaps*
- *Brain Drain*

Concept of leadership



- *Leadership is an art of influencing subordinates to direct their will, abilities and efforts for the achievements of predetermined organizational goals.*
- *Leadership is the process of guiding and influencing the behavior, activities and efforts of an individual or group for achieving the common goal.*

- *In organization, in general, by hierarchy, managers are leaders of supervisors and supervisors are the leaders of first line (operating) employees.*
- *Leader, followers and situation are the major components of leadership.*
- ***Leadership = $f(L, F, S)$***

Features of leadership



- *Leaders and Followers*
- *Ability to influence*
- *Common objectives*
- *Continues process*
- *Situational activity*



Functions of leadership

- **Goal Determination**
- **Directing**
- **Motivating**
- **Communication**
- **Supervision and Control**
- **Initiation**



Qualities of Good Leadership



- *Physical fitness*
- *A clear sense of purpose*
- *Good judgment*
- *Moral integrity*
- *Self- confidence*
- *Technical competency*
- *Managerial qualities*
- *Situational adoptability*
- *Creativity*

3.4. Leadership styles:

- Participative management,
- Management by objectives,
- management by exception,

Leadership styles

- *deals with the ways of influencing the follower's*
- *Leaders' behaviour towards subordinates*
- *Affected by objectives, personality, experience and value system, followers characteristics.*
- *Four Basic types of leadership*
 - **Autocratic,**
 - **Bureaucratic**
 - **Democratic**
 - **Free-rain (laissez faire)**
 - **Parental Leadership**



Autocratic



Laissez-faire



Democratic



1. Autocratic Leadership

- Autocratic leadership is a style where one individual holds all the decision-making power and expects subordinates to follow instructions without input or feedback.
- **Strengths:**
 - Quick decision-making.
 - Clear direction and expectations.
 - Effective in crisis or urgent situations.
- **Limitations:**
 - Discourages creativity and team input.
 - Can lower employee morale.
 - May lead to high turnover if overused.

2. Bureaucratic Leadership

- Bureaucratic leadership relies on a fixed set of rules and procedures.
- Leaders follow established policies strictly and expect others to do the same.
- **Strengths:**
 - Promotes consistency and standardization.
 - Ideal for tasks requiring accuracy and safety.
 - Minimizes favoritism and bias.
- **Limitations:**
 - Inflexible to change or innovation.
 - Slower decision-making process.
 - May demotivate employees seeking autonomy.

3. Democratic Leadership

- Democratic leadership involves team members in decision-making.
- Leaders value input and feedback before making final decisions.
- **Strengths:**
 - Encourages team participation and motivation.
 - Builds trust and stronger team relationships.
 - Enhances creativity and innovation.
- **Limitations:**
 - Slower decision-making.
 - Not ideal in emergencies.
 - May lead to conflict if consensus is not reached.

4. Free-Rein (Laissez-faire) Leadership

- Free-rein or laissez-faire leadership allows team members to work independently with minimal direction or interference from the leader.
- **Strengths:**
 - Encourages innovation and personal responsibility.
 - Suitable for highly skilled or experienced teams.
 - Fosters creativity and initiative.
- **Limitations:**
 - Lack of guidance may cause confusion.
 - Risk of low productivity or misalignment.
 - Not effective for inexperienced teams.

5. Parental Leadership

- Parental leadership is a leadership style where the leader behaves similarly to a parent—providing guidance, protection, discipline, and support to team members.
- This style combines authority with care, focusing on both performance and personal development.
- **Strengths:**
 - Builds strong emotional bonds between leader and team.
 - Encourages loyalty, respect, and trust.
 - Helps in nurturing and developing employees.
 - Provides a supportive environment, especially for new or less experienced members.
- **Limitations:**
 - May create overdependence on the leader.
 - Can blur professional boundaries if not balanced.
 - Risk of favoritism or emotional decision-making.
 - Not suitable for high-autonomy or performance-driven environments.

Servant Leadership

- **Servant Leadership (Robert K. Greenleaf, 1970)**

- Focuses on serving the needs of team members rather than the leader's personal goals.
- Emphasizes the growth, development, and well-being of employees.
- Encourages ethical behavior, empathy, and active listening.
- Builds trust, loyalty, and a supportive organizational culture.

Advantages:

- Builds trust and loyalty.
- Encourages personal and professional growth.
- Ethical and people-focused.
- Promotes a positive organizational culture.

Disadvantages:

- Slow decision-making.
- May be perceived as weak leadership.
- Risk of leader burnout.
- Not suitable for high-pressure environments.

Transformational Leadership

- **Transformational Leadership (James MacGregor Burns, 1978)**
- Inspires and motivates employees to exceed standard expectations.
- Creates and communicates a compelling vision for the organization.
- Encourages innovation, creativity, and adaptability among team members.
- Enhances commitment, morale, and overall organizational performance.

Advantages:

- Encourages innovation and creativity.
- Increases employee motivation.
- Strengthens organizational vision.
- Improves team performance.

Disadvantages:

- Risk of dependence on leader's charisma.
- Can be unrealistic in goal setting.
- May neglect short-term tasks.
- Requires continuous leader engagement.

Participative Management

- also known as **Democratic Leadership**
- This style centers on involving team members in the decision-making processes.
- Leaders actively seek input, opinions, and suggestions from their subordinates before making final decisions.



- **Key characteristics:**
 - Collaboration,
 - open communication,
 - shared responsibility,
 - empowerment of team members.

Potential benefits

- Increased employee engagement and morale,
- richer pool of ideas,
- greater buy-in to decisions,
- development of team members' skills.



Potential drawbacks

- Can be slower decision-making process,
- potential for disagreements and conflicts,
- may not be suitable for all situations (e.g., crisis situations requiring swift action).

Management by Objectives (MBO):

- developed by **Peter F. Drucker** in the 1950s.
- a strategic management approach where managers and employees collaboratively set specific, measurable goals, and work together to achieve them within a set time frame.
- focuses on setting **specific, measurable, achievable, relevant, and time-bound** (SMART) objectives collaboratively between managers and employees.
- Performance is evaluated against these agreed-upon objectives.
- **Key characteristics:**
 - Clear goal setting,
 - alignment of individual and organizational objectives,
 - autonomy in how goals are achieved,
 - performance-based evaluation.

Steps in MBO process

- Develop overall organizational goal
- Establish specific goals for various department, subunits and individuals
- Formulate action plans to reach particular goals
- Implement and maintain self-control
- Review progress periodically
- Appraise performance

• Potential benefits:

- Improved focus and direction,
- enhanced accountability,
- better communication between managers and employees,
- increased motivation through clear targets.



• Potential drawbacks:

- Can be time-consuming to set objectives,
- risk of focusing solely on measurable goals and neglecting other important aspects,
- may lead to pressure and stress if not implemented effectively.

Management by Exception (MBE):

Management by Exception is a management style in which only significant deviations from expected results or standards are brought to the attention of management.

- a leadership approach where managers focus only on significant deviations from *expected performance or standards*, rather than *overseeing every detail of operations*
- **Key characteristics:**
 - *Focus on identifying and addressing exceptions,*
 - *delegation of authority, emphasis on results,*
 - *reduced managerial involvement in routine tasks.*



• **Potential benefits:**

- Frees up managers' time to focus on strategic issues,
- empowers employees and fosters independence,
- can be efficient for well-established processes.

• **Potential drawbacks:**

- May lead to a lack of proactive problem-solving,
- potential for issues to escalate before they are addressed,
- can create a feeling of neglect if employees don't receive regular feedback or support.

3.5. Learning Organizations:



- A learning organization is a company that facilitates the continuous learning of its employees and is able to transform itself to adapt and thrive in a constantly changing environment.
- It's about fostering a culture where learning is not just an activity but an integral part of the organizational DNA.

• Key characteristics:

- **Systems Thinking:** Understanding the interconnectedness of different parts of the organization and how they influence each other.
- **Personal Mastery:** Encouraging employees to develop their skills and capabilities continuously.
- **Mental Models:** Reflecting on and challenging underlying assumptions and beliefs.
- **Shared Vision:** Creating a common understanding and commitment to the organization's goals.
- **Team Learning:** Fostering collaboration and collective intelligence within teams.



• 1. Google (USA)



- Encourages continuous learning through “20% time” where employees work on personal innovation projects.
- Provides open access to training, courses, and knowledge-sharing platforms.
- Uses data-driven learning to improve processes.

• 2. Toyota (Japan)



- Famous for the *Toyota Production System* and “Kaizen” (continuous improvement).
- Employees at all levels are trained to identify problems and suggest improvements.
- Promotes learning from mistakes and on-the-job innovation.

• 3. Apple (USA)

- Promotes learning through design thinking and rapid iteration.
- Encourages cross-functional collaboration and experimentation.
- Emphasizes learning from customer feedback and market trends.



• 4. Infosys (India)

- Has a Global Education Center in Mysuru to train thousands of engineers.
- Uses knowledge management tools and internal certification programs.
- Promotes a culture of learning, innovation, and upskilling.



• 5. Unilever (UK/Netherlands)

- Invests heavily in leadership development and digital learning platforms.
- Uses “reverse mentoring” (junior staff mentoring senior leaders).
- Learns from global teams and local markets for innovation.



Types of learning loop

- **i. single-loop learning:**

- In single-loop learning, individuals, groups, or organizations modify their actions according to the difference between expected and obtained outcomes

- **ii. Double-loop learning:**

- Double loop learning is the learning about single-loop learning.
- In double-loop learning, the entities (individuals, groups or organization) question the values, assumptions and policies that led to the actions in the first place; if they are able to view and modify those, then second-order or double-loop learning has taken place

Benefits of becoming a learning organization:



- *Increased innovation and creativity.*
- *Improved problem-solving capabilities.*
- *Greater adaptability and resilience to change.*
- *Enhanced employee engagement and retention.*
- *Competitive advantage in the marketplace.*

challenges in motivating and leading a technical workforce in the ICT industry

- **Skill Obsolescence:** Technology changes fast; existing skills rapidly become outdated, requiring constant upskilling.
- **Lack of Challenge:** Technical staff disengage quickly from routine, easy, or monotonous tasks.
- **Micromanagement:** High-skill workers demand autonomy; excessive control is a major demotivator.
- **Burnout Risk:** High-pressure deadlines and long hours lead to stress and high turnover.
- **Unseen Contributions:** Failing to recognize the value of crucial, but less visible, work (e.g., code quality, security).
- **Knowledge Silos:** Critical system knowledge gets trapped within individuals, making the organization vulnerable.
- **Divergent Goals:** Mixing staff motivated by technical mastery with those focused on management progression.
- **Remote Cohesion:** Maintaining team culture, trust, and informal communication in hybrid/remote settings.

Strategies for Motivating and Leading Technical Workforce

- **Invest in Learning:** Fund training and certifications to stop skill obsolescence.
- **Grant Autonomy:** Let teams choose how to solve problems to eliminate micromanagement.
- **Provide Challenge:** Assign complex, interesting projects to keep staff engaged.
- **Enforce Balance:** Encourage flexible hours and disconnect time to prevent burnout.
- **Reward Technical Depth:** Use metrics to recognize unseen contributions like code quality and stability.
- **Break Silos:** Mandate documentation and cross-training for effective knowledge transfer.
- **Offer Dual Paths:** Create equal progression for both Technical Experts and Managers.
- **Build Safety:** Actively foster a culture of trust and transparency for team cohesion.